NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Course Name: ETHICAL HACKING

Assignment- Week 9

TYPE OF QUESTION: MCQ/MSQ/SA

Number of questions: 10 **Total mark: 10 x 1 = 10**

QUESTION 1:

Which of the following statement(s) is/are true for sniffing?

a. It is a process of analyzing network activity by capturing network traffic.
b. It is a process of finding vulnerabilities in a network.
c. It is a process used for user enumeration.
d. It may require the NIC to operate in promiscuous mode to capture frames not specifically addressed to the host.
e. None of these.

Correct Answer: a, d

Detailed Solution: Sniffing is the process of analyzing network activity by capturing traffic. A NIC can be set to promiscuous mode to capture frames not specifically addressed to the host. It is not used to find vulnerabilities or enumerate users, which are separate techniques.

Thus, the correct options are (a) and (d).

QUESTION 2:

Which of the following is/are **not** a feature of Wireshark?

a. Capturing and analyzing network packets.
b. Generating statistical reports on captured traffic.
c. Manipulating and modifying live network traffic.
d. Displaying packet details with GUI support.
e. Carrying out SQL injection attacks.
f. Filtering captured packets by protocol.

Correct Answer: c, e

Detailed Solution: Wireshark can capture, analyze, filter, and generate statistical reports on traffic with GUI support. It cannot manipulate live traffic or perform SQL injection attacks.
Thus, the correct options are (c) and (e).

QUESTION 3:

Which of the following protocols are vulnerable to sniffing attacks?

a. HTTP
b. FTP
c. Telnet
d. HTTPS
e. SSL
f. SSH

Correct Answer: a, b, c

Detailed Solution: HTTP, FTP, and Telnet transmit data in plaintext, making them vulnerable to sniffing. HTTPS, SSL, and SSH encrypt data in transit, protecting it from being read even if intercepted.
Thus, the correct options are (a), (b), and (c).

QUESTION 4:

Which of the following feature(s) is/are **not** present in the Ettercap tool?

a. IP-based and MAC-based filtering.
b. Character injection into live connections.
c. Packet filtering and dropping.
d. SQL injection automation.
e. Password cracking via dictionary attack.

Correct Answer: d, e

Detailed Solution: Ettercap supports IP/MAC-based filtering, character injection into active connections, and packet filtering/dropping. It does not automate SQL injection or perform dictionary-based password cracking.

Thus, the correct options are (d) and (e).

QUESTION 5:

Which of the following is **not** a human-based social engineering attack?

a. Impersonation
b. Shoulder surfing
c. Piggybacking/tailgating
d. Phishing email

Correct Answer: d

Detailed Solution: Impersonation, shoulder surfing, and piggybacking/tailgating all involve direct human interaction or physical presence, so they are human-based social engineering. Phishing is a computer-based social engineering attack, carried out through electronic communication rather than direct human contact.

Thus, the correct option is (d).

QUESTION 6:

A DoS attack that exploits the TCP three-way handshake by leaving connections incomplete is called:

a. SYN Flooding
b. ICMP Flood
c. Ping of Death
d. UDP Flood

Correct Answer: a

Detailed Solution: SYN Flooding exploits the TCP handshake by sending repeated SYN requests without completing the final ACK step, leaving many half-open connections that exhaust the server's connection queue.

Thus, the correct option is (a).

QUESTION 7:

Which tool is specifically known for performing DoS attacks by sending partial HTTP requests and never completing them?

a. Hydra
b. Wireshark
c. Slowloris
d. Crunch

Correct Answer: c

Detailed Solution: Slowloris opens many connections to a web server and sends partial HTTP requests, keeping them open indefinitely to exhaust the server's available connections and cause a denial of service.

Thus, the correct option is (c).

QUESTION 8:

Which mode allows a network interface card (NIC) to capture frames that are not specifically addressed to the host?

a. Monitor mode
b. Transparent mode
c. Promiscuous mode
d. Bridge mode

Correct Answer: c

Detailed Solution: Promiscuous mode allows the NIC to capture frames that are not specifically addressed to its own MAC address.

Thus, the correct option is (c).

QUESTION 9:

Which of the following is/are examples of computer-based social engineering attacks?

a. Phishing emails
b. Pop-up window scams
c. Shoulder surfing
d. Chain letter emails
e. Tailgating

Correct Answer: a, b, d

Detailed Solution: Phishing emails, pop-up window scams, and chain letter emails are delivered through computer/electronic means, classifying them as computer-based social engineering. Shoulder surfing and tailgating require physical presence, making them human-based social engineering attacks.

Thus, the correct options are (a), (b), and (d).

QUESTION 10:

What is the primary purpose of ARP spoofing in the context of sniffing attacks?

a. To increase the bandwidth available to the attacker.
b. To redirect traffic between two hosts through the attacker's machine for interception.
c. To encrypt network traffic for secure transmission.
d. To permanently disable a target host's network interface.

Correct Answer: b

Detailed Solution: ARP spoofing associates the attacker's MAC address with a legitimate host's IP address, redirecting traffic intended for that host through the attacker's machine and enabling interception (man-in-the-middle), often used to facilitate sniffing on switched networks.

Thus, the correct option is (b).

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